COMP 233
Discrete Mathematics with Applications (Epp, 4th Edition)

Lecture 3.2 – Predicates and Quantified Statements II

Negation • Universal Conditionals • Quantifiers and AND/OR • Vacuous Truth • Related Conditionals • Necessary and Sufficient Conditions

Chapter 3
The Logic of Quantified Statements
Section 3.2 (pp. 108–116)

1 Quantifiers and Predicates (Quick Review)

Predicate

A **predicate** $P(x)$ is a statement that can be true or false for each element x in a domain D.

Quantifiers

Universal Quantifier ($\forall$)

$\forall x \in D, P(x)$
means "$P(x)$ is true for all x in D."

All elements satisfy $P(x)$.

Existential Quantifier ($\exists$)

$\exists x \in D, P(x)$
means "$P(x)$ is true for at least one x in D."

At least one element satisfies $P(x)$.

2 Negation of Quantified Statements

Negation Rules

Negation of a Universal Statement

$$\neg(\forall x \in D, P(x)) \equiv \exists x \in D, \neg P(x)$$

To show a universal statement is false, find one **counterexample** for which $P(x)$ is false.

Negation of an Existential Statement

$$\neg(\exists x \in D, P(x)) \equiv \forall x \in D, \neg P(x)$$

To show an existential statement is false, show that **no** element satisfies $P(x)$.

Example 1 (Universal)

$P(x)$: x is a multiple of 3, $D = \{1, 2, 3, 4\}$

$\neg(\forall x \in D, P(x)) \equiv \exists x \in D, \neg P(x)$

The statement is false because $x = 1$ is not a multiple of 3.

Example 2 (Existential)

$P(x)$: $x > 5$, $D = \{1, 2, 3\}$

$\neg(\exists x \in D, P(x)) \equiv \forall x \in D, \neg P(x)$

The negation is true because no element in D is greater than 5.

3 Negation of a Universal Conditional

Statement: $\forall x \in D, P(x) \to Q(x)$

Derivation of the Negation

$$\neg(\forall x\,(P(x) \to Q(x))) \equiv \exists x\,\neg(P(x) \to Q(x)) \equiv \exists x\,(P(x) \wedge \neg Q(x))$$

To show that a universal conditional is false, find one element for which $P(x)$ is true and $Q(x)$ is false.

Example: Program verification

$P(x)$: program x compiles successfully
$Q(x)$: program x is bug-free

$$\neg(\forall x\,(P(x) \to Q(x))) \equiv \exists x\,(P(x) \wedge \neg Q(x))$$

A single program that compiles successfully and contains a bug proves the statement is false.

4 Quantifiers as AND and OR (Finite Domains)

If the domain D is finite, quantifiers can be written using logical connectives.

Universal as AND

$D = \{1, 2, 3\}$
$\forall x \in D, P(x)$
$\equiv P(1) \wedge P(2) \wedge P(3)$

All individual statements must be true.

Existential as OR

$D = \{1, 2, 3\}$
$\exists x \in D, P(x)$
$\equiv P(1) \vee P(2) \vee P(3)$

At least **one** individual statement must be true.

Example

Let $D = \{1, 2, 3\}$ and $P(x)$: "x is even."

- $\forall x \in D, P(x) \equiv P(1) \wedge P(2) \wedge P(3) = F$ (since $P(1)$ is false)
- $\exists x \in D, P(x) \equiv P(1) \vee P(2) \vee P(3) = T$ (since $P(2)$ is true)

5 Vacuous Truth

A universal conditional $\forall x\,(P(x) \to Q(x))$ is vacuously true if there is no element in the domain for which $P(x)$ is true.

Case 1: No element satisfies P(x)

If $P(x)$ is false for all x in D, then $\forall x\,(P(x) \to Q(x))$ is true.

There is no counterexample.

Case 2: False antecedent for all x

If $P(x)$ is false for every x in D, then each conditional $P(x) \to Q(x)$ is true, and so the universal conditional is true.

Example 1: Empty domain

Let $D = \emptyset$.
$\forall x \in D, P(x)$ is true (vacuously) because there is no element that violates $P(x)$.

Example 2: No students in class

$D = \{x \in \mathbb{R} \mid x < 0\}$ and $P(x)$: $x \geq 0$.

$\forall x \in D\,(P(x) \to Q(x))$ is true because $P(x)$ is never true for any x in D.

6 Converse, Inverse, and Contrapositive

For a statement of the form $\forall x\,(P(x) \to Q(x))$, consider the related forms:

Form	Symbolic expression	Relation to original
Original	$\forall x\,(P(x) \to Q(x))$	—
Converse	$\forall x\,(Q(x) \to P(x))$	Not logically equivalent in general.
Inverse	$\forall x\,(\neg P(x) \to \neg Q(x))$	Logically equivalent to converse.
Contrapositive	$\forall x\,(\neg Q(x) \to \neg P(x))$	Logically equivalent to original.

Example: Student enrollment

$P(x)$: x is enrolled in COMP233 $Q(x)$: x is a university student

- **Original:** Every student enrolled in COMP233 is a university student.
- **Converse:** Every university student is enrolled in COMP233. (Not equivalent)
- **Inverse:** If a student is not enrolled in COMP233, then the student is not a university student. (Equivalent to converse)
- **Contrapositive:** If a student is not a university student, then the student is not enrolled in COMP233. (Equivalent to original)

7 Necessary and Sufficient Conditions

For statements about properties $P(x)$ and $Q(x)$:

English statement	Symbolic form
$P(x)$ is sufficient for $Q(x)$	$P(x) \to Q(x)$
$P(x)$ is necessary for $Q(x)$	$Q(x) \to P(x)$
$P(x)$ is necessary and sufficient for $Q(x)$	$P(x) \leftrightarrow Q(x)$
$P(x)$ only if $Q(x)$	$P(x) \to Q(x)$
$P(x)$ if $Q(x)$	$Q(x) \to P(x)$

Example: Administrator access

$P(x)$: x is an administrator
$Q(x)$: x has access to the dashboard

- If a user is an administrator, then the user has access to the dashboard. $P(x) \to Q(x)$
- Being an administrator is sufficient for dashboard access.
- Having dashboard access is necessary for being an administrator: $Q(x) \to P(x)$
- The converse need not be true (a non-admin may have access).

8 Examples and Counterexamples

Example 1: Universal statement

Statement: $\forall x \in \mathbb{Z}, x^2 \geq 0$
Truth value: True
Reason: The square of every integer is non-negative.

Counterexample 1

Statement: $\forall x \in \mathbb{Z}, x > 0$
Truth value: False
Counterexample: $x = 0$ (and also any negative integer).

Example 2: Existential statement

Statement: $\exists x \in \mathbb{Z}, x^2 = 4$
Truth value: True
Witness: $x = 2$ (also $x = -2$).

Counterexample 2

Statement: $\exists x \in \mathbb{Z}, x^2 = -1$
Truth value: False
Reason: No integer has square -1.

9 Summary: Key Ideas and Connections

1. Negating a quantified statement changes the quantifier and negates the predicate.
2. A universal statement is false if there exists one counterexample.
3. A universal conditional is false if there exists one element with $P(x)$ true and $Q(x)$ false.
4. Over a finite domain, $\forall$ behaves like AND and $\exists$ behaves like OR.
5. A universal conditional can be vacuously true when no element satisfies the antecedent.
6. The contrapositive of a conditional is logically equivalent to the original statement.
7. P is sufficient for Q means $P \to Q$; P is necessary for Q means $Q \to P$.
8. Distinguish carefully between negation, converse, inverse, and contrapositive.